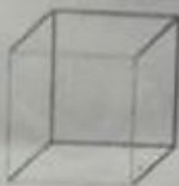


Marks: 70

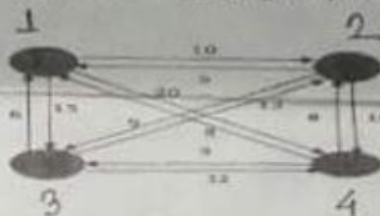
[Answer any five questions. Assume reasonable value for any missing data]

Time: 1.00 Hours

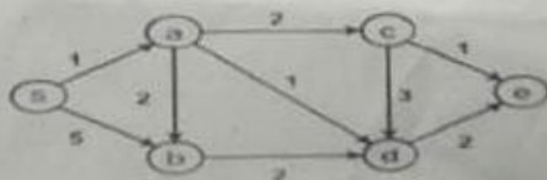
1. a) What is Backtracking? Write down The general steps of backtracking. 4
- b) Write down the Characteristics of Huffman Coding. 4
- c) Draw Six Hamiltonian cycle from the Graph which given below. 6



2. a) Define greedy method. Write down the basic algorithm for greedy approach. 4
- b) What is Travelling Sales Person (TSP)? Solve the following TSP problem for the given matrix and graph (assume root is 1): 5



- c) Write a greedy algorithm for single source shortest path problem. Based on this algorithm find the single source shortest path for the following graph: 5



3. a) Consider the following 9 activities with their starting and finishing time. 7

Activity: (A₁, A₂, A₃, A₄, A₅, A₆, A₇, A₈, A₉)

Start: (1, 2, 3, 4, 5, 6, 7, 8, 9)

Finish: (3, 5, 7, 8, 9, 11, 14, 16)

Compute a schedule where the greatest number of activities takes place Using Activity selection Problem.

- b) Write down the Real-life Applications of Activity Selection Problem. 3
- c) Write down the GREEDY ACTIVITY SELECTOR Algorithm. 4

4. a) Define and explain i) DAG ii) indegree iii) outdegree. 3
- b) What is parallel algorithm? Explain Flynn's Classification of computers with required Diagram. 6
- c) Write down the steps of Interpolation search. 2
- d) Write down the pseudo-code of Graph coloring Problem. 3

A₁, A₂, A₃, A₄, A₅